SRIVATSAV SARAVANAN

Software Engineer
srivatsavs98@gmail.com | +47 96705697 | LinkedIn: srivatsav-s | GitHub: Srivatsav1298 | Portfolio: srivatsav
Oslo, Norway

PROFILE SUMMARY

I am a passionnate Software Engineer with 3+ years of experience building data-intensive applications. Strong foundation in Python, SQL, and cloud platforms(AWS), with hands-on experience delivering AI-native and LLM-based solutions.

EDUCATION

Master's in Data Science, Norwegian University of Life Sciences (NMBU), Ås AUG 2024 - JUN 2026
Relevant Coursework: Data-to-Decision, SQL for Data Handling and Analysis, Python Programming

Bachelor's in Computer Science and Engineering, India AUG 2016 - APR 2020
Relevant Coursework: Software Engineering, Web Development, System Design

SKILLS

- **Programming & Backend**: Python, FastAPI, Pandas
- **Databases:** SQL**,** PostgreSQL
- **Data Engineering:** PySpark, Data Modelling, DBT
- **Data and AI:** Machine Learning, LangChain, LLM APIs (Claude, LLaMA)
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Git
- Frontend**:** React, JavaScript, HTML, CSS

WORK EXPERIENCE

Teaching Assistant | NMBU | DBMS SEPT 2025 - DEC 2025

- Mentored 40+ students in SQL optimization and database design, improving their ability to design efficient schemas and write performant queries for real-world datasets.
- Developed hands-on exercises that simulated real-world database challenges, enhancing student learning outcomes.

ORION INNOVATION | Software Engineer | .NET, SQL, Angular, TypeScript, HTML, CSS NOV 2020 - JUL 2024

- Employee Onboarding Platform: Developed and converted an internal onboarding application into a multi-device responsive web platform, supporting 4,000+ users and increasing onboarding completion and user activity by 13%. Collaborated with UX designers to ensure seamless user experience across desktop and mobile devices.
- Application Migration - Tennis Australia: Contributed to the migration of Tennis Australia's application infrastructure from external hosting to an in-house environment, reducing operational costs by 30% and improving system control, security, and reliability. Worked closely with DevOps team to ensure zero-downtime deployment.
- Database Performance Optimization: Identified and resolved leaves performance bottlenecks in employee attendance system by optimizing SQL queries and implementing strategic indexing, improving system reliability by approximately 35%. Reduced query execution times from 45 seconds to under 3 seconds through systematic analysis and testing.
- Agile Team Collaboration: Worked closely with cross-functional Agile teams (developers, QA, product owners) through sprint planning, daily stand-ups, and retrospectives, enabling faster and more predictable feature delivery (approximately 20% improvement). Actively participated in code reviews and knowledge-sharing sessions.

PROJECTS

Thesis JAN 2026 - PRESENT

Explainable Financial Reasoning under Data Constraints: An Empirical Study Using Norwegian Data

- Designed and implemented a production-grade ETL pipeline processing 10,000+ Norwegian household financial records from Statistics Norway (SSB).
- Designed a knowledge graph (47 nodes, 66 edges) to validate and trace AI reasoning paths in financial data.
- Developed an asynchronous evaluation framework using FastAPI for automated crash recovery.
- Discovered critical "coverage-dependent overconfidence" in LLMs, highlighting AI safety in financial contexts.
- Tech Stack: Python, ETL Pipeline, PostgreSQL, FastAPI, RDFLib, Pandas, SciPy, LLM APIs

Financial AI Agent | Python, Llama, Statistics Norway API (SSB) | Source Code SEPT 2025 - NOV 2025

- Built an AI-assisted backend service for answering Norwegian financial and economic queries using real public datasets.
- Focused on explainable outputs and reliable data retrieval rather than black-box predictions.

Analytics Dashboard | Python, PySpark, MongoDB, Streamlit - Website Link SEPT 2025 - NOV 2025

- Engineered a scalable ETL pipeline using PySpark and AWS to process 1M+ hourly electricity consumption records from Elhub Norway, ensuring 99.9% data consistency for downstream consumers.
- Optimized data storage and retrieval using MongoDB, reducing query response times for real-time monitoring and operational decision-making.

AV Danse Studio| AWS, JavaScript, HTML, CSS | Visit Website JAN 2025 - APR 2025

- Developed a digital platform for a dance center in Norway using AWS (S3, CloudFront, Route 53), achieving 99.99% uptime and a 50% reduction in operational costs through a full migration from legacy hosting.
- Implemented a fully automated CI/CD pipeline that doubled deployment velocity and supported a 2x increase in concurrent user traffic without performance degradation.

Resume Analyzer| React, TypeScript, Claude - 3.5 - Sonnet | Visit Website SEPT 2025 - NOV 2025

- Created an AI-driven feedback engine using TypeScript and LangChain, implementing structured prompting and output parsing to ensure high-precision data extraction from unstructured text.
- Improved recommendation accuracy by 40% by designing a validation layer to cross-reference AI outputs with industry datasets, while optimizing API latency through strategic caching and token-usage management.

CERTIFICATIONS & AWARDS

- Databricks Fundamentals FEB 2026
- 3rd Place: GenAI Hackathon - AWS Norway NOV 2025
- Cloud Computing Badge - AWS NOV 2025
- GenAI Concepts Certification - DataCamp FEB 2025
- Star of the Month (2x) - Orion Innovation JUL 2022, NOV 2023

LANGUAGES

- English - Fluent
- Norwegian - B1 (Actively Improving)

HOBBIES

- Skiing
- Cricket